Abdullah Emad Eldeen

AI & Embedded Systems Student | Machine Learning | TinyML | IoT

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Profile

Final-year Artificial Intelligence undergraduate (Intelligent Systems track, Menoufia University) with hands-on experience shipping embedded AI systems, from TinyML keyword spotting on ESP32-S3 to full-stack computer vision applications. Comfortable across the stack: model training in TensorFlow/PyTorch, on-device deployment with Edge Impulse and TensorFlow Lite Micro, and backend integration with FastAPI and MQTT. Seeking an AI/ML, Embedded AI, or IoT internship.

Education

B.Sc. in Artificial Intelligence, Intelligent Systems Department

Faculty of Artificial Intelligence, Menoufia University | 09/2023 - Present (Expected Graduation: 2027)

Technical Skills

- Programming: Python, C, C++
- Machine Learning: Supervised & Unsupervised Learning, Model Evaluation, Dimensionality Reduction (PCA), Feature Engineering
- Deep Learning: Neural Networks, CNNs, Transfer Learning, Model Optimization & Quantization
- Computer Vision: OpenCV, MediaPipe, YOLO, Image Classification, Face Recognition
- Embedded AI / IoT: ESP32, Arduino, Microcontrollers, TinyML, MQTT, I2S, Sensors
- Frameworks / Tools: TensorFlow, PyTorch, NumPy, Pandas, Edge Impulse, TensorFlow Lite Micro, Arduino IDE, PlatformIO, ESP-IDF, FastAPI

Projects

ESP32-S3 Voice-Controlled Keyword Spotting (TinyML)

ESP32-S3 | INMP441 | Edge Impulse | TensorFlow Lite Micro (MobileNetV1, INT8)

- Built a real-time, fully on-device voice-command recognition pipeline on an ESP32-S3 using an INMP441 I2S microphone and an Edge Impulse MobileNetV1 transfer-learning model, achieving 95.5% validation accuracy across 3 keyword classes on an 11,570-sample dataset.
- Optimized the quantized (INT8) model for embedded deployment, reaching 271ms inference time with only 36.5KB peak RAM and 109.5KB flash usage on-device.
- Implemented 16kHz I2S audio acquisition and MFE (Mel-Frequency Energy) preprocessing, resolving audio formatting and bit-alignment inconsistencies between training and embedded environments.

Smart Adaptive Environment Monitoring System

ESP32-S3 | MQTT | Python | PCA, SOM, RBF, Fuzzy Logic, Reinforcement Learning

- Designed a closed-loop IoT + AI system fusing ESP32-S3 sensor hardware (DHT22, MQ-135, LDR) with a 6-stage host-side AI inference pipeline (PCA, SOM, RBF Network, Fuzzy Logic, Q-Learning, ART1/ART2) communicating over MQTT.
- Trained a 40-center RBF multi-output classifier and a 20x20 Self-Organizing Map, and used a Genetic Algorithm (population 20, 20 generations) to optimize RBF weights and fuzzy rule parameters.
- Built dual real-time dashboards (Flask + Socket.IO, Streamlit) with safety-override logic so hazardous readings always trigger maximum actuator response regardless of the RL agent's output.

Real-Time ASL Alphabet Recognition

Python | TensorFlow/Keras | MediaPipe | Computer Vision

- Built a real-time American Sign Language alphabet recognizer using MediaPipe hand tracking (21 3D landmarks -> 63 features) and a custom-trained TensorFlow/Keras classifier.

- Applied wrist-relative normalization and scale-invariant feature scaling so recognition stays robust across different hand sizes and camera distances.
- Added a 5-frame rolling-average smoothing filter to stabilize predictions and reduce flicker, and built a Tkinter GUI to compose words/sentences live from recognized letters.

Celebrity Face Recognition System

Python | FaceNet | ArcFace | FastAPI | HTML/CSS/JavaScript

- Developed a full-stack facial recognition app recognizing 105 celebrity classes, combining MTCNN face detection with FaceNet (InceptionResnetV1) embeddings (512-dim, 160x160 input) fine-tuned using ArcFace (Additive Angular Margin) loss.
- Built a FastAPI backend serving top-3 ranked predictions with confidence scoring, filtering out low-confidence (<85%) matches to reduce false positives.
- Built a JavaScript/HTML frontend for interactive image upload, live health-check status, and celebrity search.

Training & Courses

- Generative AI Engineering with LLMs - IBM / Coursera (2026)
- Deep Learning with PyTorch - IBM / Coursera
- Fundamentals of TinyML - HarvardX / edX
- TinyML2: Applications of TinyML - HarvardX / edX
- Introduction to Embedded Machine Learning - Edge Impulse (2026)
- Machine Learning Engineer Certification Preparation - Google Cloud / Coursera

Languages

Arabic - Native | English - Professional Working Proficiency | German - Basic